

Fed-RIS: Distributed Phase Optimization for Reconfigurable Intelligent Surfaces via On-Chip Federated Learning

Anonymous Author(s)
Affiliation withheld for review

Abstract—Large reconfigurable intelligent surfaces (RIS) need instantaneous channel-state information (CSI) to set their phases, yet pooling every element’s CSI at a central controller and solving an iterative optimization in each coherence block scales poorly. We propose Fed-RIS, which federates phase optimization across the tiles of a single surface. A 1024-element RIS is split into 16 tiles of 8×8 elements; each tile trains a shared graph-attention phase predictor on its own CSI with a differentiable sum-rate objective, and only INT8 model deltas cross an on-chip Network-on-Chip (NoC)—raw CSI never leaves a tile, and inference is one forward pass per block. On identical 28 GHz channels with a 30 dB-blocked direct path, Fed-RIS reaches BER 10^{-3} 36.7 dB earlier than the blocked link, saves 31.4 dB of transmit power at 4 bit/s/Hz, and lies within 1.3 dB of the perfect-CSI bound. Under imperfect CSI it degrades least of all designs (3.2 dB vs. 5.1 dB for AO/SCA), 2-bit phase shifters cost only 0.82 dB in agreement with theory, and a Torus NoC with RingAllReduce cuts aggregation latency $6.4\times$ and energy by 56%.

Index Terms—Reconfigurable intelligent surface, federated learning, graph attention network, network-on-chip, mmWave.

I. Introduction

Reconfigurable intelligent surfaces (RIS) reshape the radio channel with programmable, nearly passive reflecting elements [1], [2]. At mmWave, where blockage can attenuate the direct path by tens of dB [3], an RIS offers a reflected route whose coherent gain grows as $O(N^2)$ [4]. Achieving it requires aligning every element with its cascaded channel.

Model-based phase design—alternating optimization (AO) [4], SCA [6], ADMM [7] and SDR [8]—needs full CSI at a central controller and an iterative solve in every coherence block (2.0–21.9 ms in our measurements). Learned designs based on DRL [9] and GNNs [10], [11] reduce inference to a single forward pass but are trained on CSI pooled centrally. Federated learning (FL) [13]–[15] has been applied with users or devices as clients [16], [17], whose outputs are independent and whose communication is counted in abstract bytes. A modular RIS, however, is itself a distributed system: each tile estimates only its own CSI slice and is linked to its peers by an on-chip interconnect [18], [19].

Contributions. (C1) Fed-RIS is, to our knowledge, the first framework to federate phase optimization across the tiles of one metasurface; the clients’ outputs are physically

summed, so training is per tile but scoring is over the full aperture. (C2) A link-quality objective: a differentiable, genie-normalized sum-rate with a (cos, sin) output head, replacing MSE regression onto phase labels. (C3) Hardware in the loop: quantization, jitter and normalized CSI error are applied before scoring and validated against the sinc bound. (C4) An interconnect-aware cost model on a routed NoC, with a closed-form crossover against raw-CSI upload. (C5) A link-level benchmark of seven schemes on identical channel realizations.

II. System Model

A single-antenna 28 GHz BS ($P_t = 30$ dBm) serves a user through an RIS of $N = 1024$ elements arranged as $T = 16$ tiles (4×4) of $N_t = 64$ elements (8×8). The direct path suffers 30 dB blockage [3]; noise is $\sigma^2 = -90$ dBm. The effective channel and received SNR are

$$h_{\text{eff}} = h_d + \sum_{t=1}^T \sum_{n=1}^{N_t} c_{t,n} e^{j\theta_{t,n}}, \quad \gamma = \rho |h_{\text{eff}}|^2, \quad (1)$$

where $c_{t,n}$ is the cascaded BS–RIS–user coefficient of element n in tile t and $\rho = P_t/\sigma^2$ is the transmit SNR. Links are spatially correlated Rician ($K = 10$ dB, correlation 0.7, 5 NLoS paths) with path-loss exponents 2.5 (RIS hops) and 3.5 (direct) and per-element aperture gain $4\pi A_e/\lambda^2$. The rate-optimal design is the MRC alignment $\theta_{t,n}^* = \angle h_d - \angle c_{t,n}$, which needs perfect global CSI and serves as an oracle bound. Fed-RIS instead learns a shared mapping $\mathbf{f}_w : \hat{\mathbf{c}}_t \mapsto \boldsymbol{\theta}_t$ from each tile’s noisy local estimate, without pooling CSI.

III. The Fed-RIS Framework

Features and predictor. Each tile uses the real and imaginary parts of its cascaded and direct coefficients with per-block RMS normalization. The per-sample global phase $\angle h_d$ is removed from the target and re-added at application (Fig. 1C); otherwise it swamps the per-element structure (label std 1.850 rad vs. 1.814 rad for pure noise). The predictor is a 3-layer graph attention network [12] over the 4-neighbour 8×8 element grid (8 heads, width 256, per-element cascade projections, 315,906 parameters). It outputs $(\cos \theta_n, \sin \theta_n)$, avoiding the $0/2\pi$ wrap-around discontinuity.

Federated phase control for a tiled RIS: end-to-end system

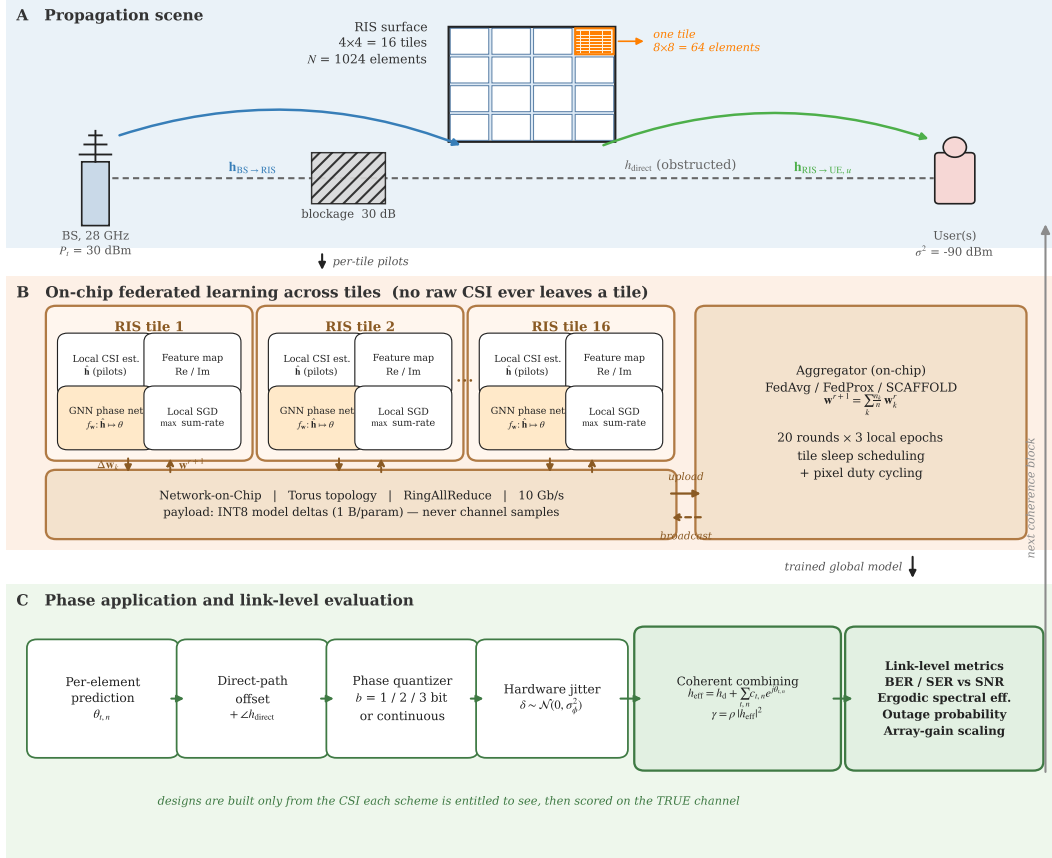


Fig. 1. Fed-RIS architecture. (A) 28 GHz scene with blocked direct path and a 4×4 -tile, 1024-element RIS. (B) Each tile trains the shared GAT predictor on local CSI; only INT8 deltas cross the Torus NoC (RingAllReduce). (C) Phases are offset by $\angle h_d$, quantized, jittered, and scored on the true channel.

Objective. Each tile minimizes

$$\mathcal{L} = 1 - \frac{\sum_u w_u \log_2(1 + \text{SINR}_u)}{R_{\text{genie}}}, \quad (2)$$

computed from (1) with real arithmetic and normalized by the batch MRC rate R_{genie} so that gradients do not vanish when $\text{SINR} \ll 1$.

Federation over the NoC. In each of $R = 20$ rounds, every tile runs $E = 3$ local Adam epochs ($\text{lr } 3 \times 10^{-4}$, batch 64) and sends its INT8-quantized update; FedAvg [13] forms $\mathbf{w}^{r+1} = \mathbf{w}^r + \sum_t \frac{n_t}{n} \Delta \mathbf{w}_t$ via RingAllReduce [20] on a 10 Gb/s 2D Torus. At run time each tile computes $\theta_t = f_w(\hat{\mathbf{c}}_t) + \angle h_d$ in one $O(N)$ forward pass.

Impairments. Predicted phases undergo b -bit quantization (theoretical loss $20 \log_{10} \text{sinc}(2^{-b})$ [5]), Gaussian jitter σ_ϕ , and CSI error $\hat{\mathbf{c}} = \mathbf{c} + \mathbf{e}$, $\mathbf{e} \sim \mathcal{CN}(\mathbf{0}, \epsilon \mathbb{E}[\mathbf{c}^2] \mathbf{I})$. Normalizing by channel power is essential: with $|\mathbf{c}| \sim 10^{-10}$, an absolute variance erases the estimate, whereas $\epsilon \in [10^{-3}, 1]$ spans 1.7° – 45° of phase error.

Cost crossover. FL moves $2RP$ INT8 bytes per tile over training, while centralized training uploads $4SD$ FP32

bytes of raw CSI (S samples of dimension D). Federation is cheaper only when

$$S > S^* = 2RP/(4D) \approx 23,752 \text{ samples/tile} \quad (3)$$

for $R = 20$, $P = 315,906$ and $D = 133$.

IV. Evaluation Setup

All schemes design phases from the CSI they are entitled to and are scored on the same 600 true channel realizations (seed 42); the solvers use the exact noisy estimate the learners see. Baselines: no RIS; random phases; AO [4]; SCA [6]; Centralized DL, the identical network, loss and schedule trained on pooled data, so that it differs from Fed-RIS only in data locality; and the perfect-CSI MRC bound. Fed-RIS uses 600 training samples per tile. BER is computed semi-analytically (exact conditional AWGN error averaged over realizations) and cross-checked against symbol-level Monte-Carlo simulation.

V. Results

BER, rate and outage. Table I and Fig. 2 summarize the link-level results. Fed-RIS reaches $\text{BER } 10^{-3}$ (QPSK) at

TABLE I
Link-Level Comparison on 600 Identical Channel Realizations (1024 Elements, 28 GHz, 30 dB Blockage). ρ : Transmit SNR Needed to Reach the Target (Lower Is Better).

Scheme	CSI stays local	Per-block cost	Array gain (dB)	$\rho@10^{-3}$ QPSK (dB)	$\rho@10^{-3}$ 16-QAM (dB)	SE @ $\rho=99$ dB (bit/s/Hz)	Power saving @4 b/s/Hz (dB)	Gap to Fed-RIS (dB)
No RIS (blocked)	✓	—	0.00	135.9	—	0.16	0.0	+36.7
Random phases	✓	none	0.65	136.9	—	0.35	7.2	+37.7
AO [4]	×	21.9 ms solve	14.80	98.0	104.6	5.23	33.2	-1.2
SCA [6]	×	2.0 ms solve	14.85	98.0	104.6	5.24	33.3	-1.2
Centralized DL	×	1 fwd pass	14.72	98.1	104.7	5.19	33.1	-1.1
Fed-RIS (proposed)	✓	1 fwd pass	12.88	99.2	105.8	4.64	31.4	0.0
Perfect-CSI MRC (bound)	—	closed form	14.91	97.9	104.5	5.26	33.3	-1.3

“—” in the ρ columns: target not reached within $\rho \in [88, 142]$ dB. Array gain is relative to the blocked direct link.

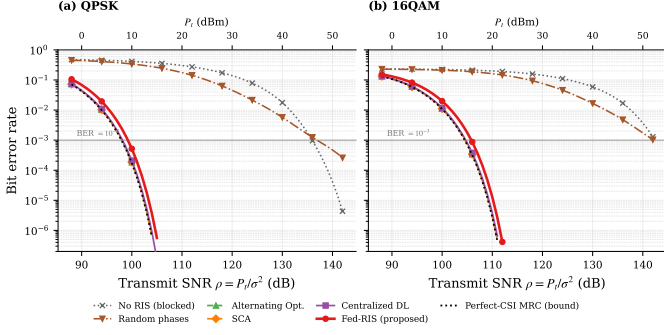


Fig. 2. BER vs. transmit SNR for QPSK and 16-QAM. Fed-RIS recovers 36.7 dB over the blocked link and lies 1.3 dB from the MRC bound.

$\rho = 99.2$ dB, 36.7 dB before the blocked link and 37.7 dB before random phases, and only 1.3 dB behind the perfect-CSI bound. The same 1.3 dB gap holds for 16-QAM, which the blocked link and random phases never reach. The 1.1 dB gap to Centralized DL is exactly the price of keeping CSI on-tile. At $\rho = 99$ dB, Fed-RIS delivers 4.64 bit/s/Hz, 29 $\times$ the blocked link and 88% of the oracle rate, and at a 4 bit/s/Hz target it saves 31.4 dB of transmit power (oracle: 33.3 dB; random phases: 7.2 dB). For a 1 bit/s/Hz outage threshold at $\rho = 95$ dB, the blocked link is in outage 98.2% of the time, whereas Fed-RIS shows zero outage across all 600 realizations.

Robustness to imperfect CSI. Solvers treat the noisy estimate as exact, so they degrade fastest (Table IIa, Fig. 3). Over $\epsilon = 0 \rightarrow 1$, AO and SCA lose 5.1 dB of received SNR, Centralized DL 4.6 dB, and Fed-RIS only 3.2 dB. Its 1.9 dB perfect-CSI gap to AO thus closes completely at $\epsilon = 1$ (13.0 vs. 13.0 dB), where its BER is also lower (1.09×10^{-2} vs. 1.13×10^{-2}). The learned predictor carries a channel prior that regularizes its response to estimation noise.

Hardware impairments. The measured quantization losses track the sinc bound (Table IIb), so the simulation is falsifiable against theory. 2-bit shifters are the practical knee (0.82 dB). Jitter is second-order: 0.03 dB at 5° and 1.07 dB at 30°.

Array-gain scaling. From $N = 256$ to 512 all designs

TABLE II
Robustness, Hardware and Scaling (Mean Received SNR / Loss in dB at $\rho = 99$ dB)

(a) Normalized CSI error ϵ						
Scheme	0	0.01	0.1	0.3	1	Loss
AO / SCA	18.1/18.2	18.1	17.5	16.3	13.0/13.1	5.1
Centralized DL	18.0	18.0	17.6	16.5	13.4	4.6
Fed-RIS	16.2	16.2	15.8	15.1	13.0	3.2

(b) Phase quantization loss			
	1-bit	2-bit	3-bit
Theory $20\log_{10}\text{sinc}(2^{-b})$	-3.92	-0.91	-0.22
MRC (measured)	-3.40	-0.84	-0.21
Fed-RIS (measured)	-3.26	-0.82	-0.20

(c) RMS phase jitter loss, Fed-RIS			
	5°: -0.03	15°: -0.27	30°: -1.07

(d) Array-gain scaling, active elements N					
	64	256	512	768	1024
MRC (bound)	3.8	10.5	16.9	17.6	18.2
Fed-RIS	3.6	8.9	14.7	15.6	16.2
Ideal N^2 law	3.8	15.8	21.8	25.4	27.9

TABLE III
Learning-System and On-Chip Cost (16 Tiles, 20 Rounds)

(a) Predictor ablation (system-level suite; relative ranking only)				
Model	Params	FL traffic (MB)	SNR (dB)	
MLP	198,784	121.3	7.72	
CNN + SE attention	130,114	79.4	8.08	
Transformer	2,225,410	1358.3	8.28	
GAT (proposed)	315,906	192.8	8.75	
(b) NoC aggregation over full training				
Topology	Protocol	Bytes (MB)	Latency (ms)	Energy (μ J)
Mesh	ParamServer	189.5	121.3	51.2
Torus	ParamServer	189.5	80.9	38.8
Torus	AllReduce	404.4	40.4	58.2
Torus	Gossip	505.4	65.7	114.0
Torus	RingAllReduce	189.5	18.96	22.7
(c) Traffic accounting (600 samples/tile)				
FL (INT8 deltas) / raw-CSI upload		192.8 MB / 4.9 MB (39.6 $\times$)		
Crossover S^* , Eq. (3)		$\approx 23,752$ samples/tile		

follow the N^2 law (MRC: +6.4 dB). Beyond 512 elements the gain saturates for every design, including the oracle (+1.3 dB vs. +6 dB predicted), because peripheral tiles see much weaker cascaded channels (Table II d). This points to tile placement as the lever. The gap of Fed-RIS to MRC grows from 0.1 dB at $N = 64$ to 2.0 dB at 1024, identifying cross-tile coherence as the main remaining headroom.

Learning system and NoC. In the full-scale system-level suite (Table IIIa), GAT outperforms MLP by 1.0 dB

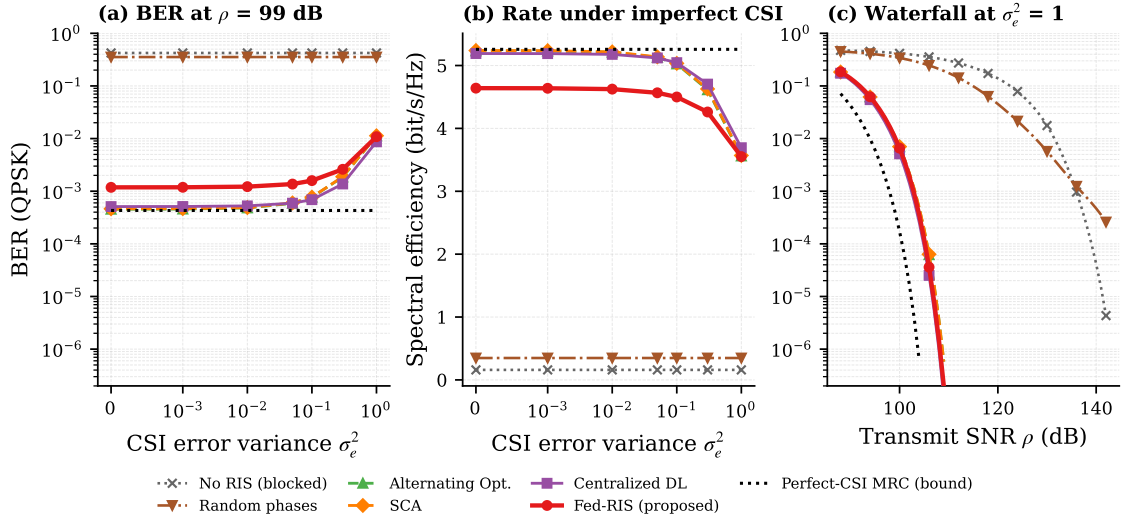


Fig. 3. Robustness to CSI error: (a) BER at $\rho = 99$ dB, (b) spectral efficiency, (c) BER waterfall at $\epsilon = 1$. Fed-RIS loses 3.2 dB over the sweep, vs. 5.1 dB for AO and SCA.

and a Transformer with $7\times$ more parameters by 0.5 dB. FedAvg, FedProx and SCAFFOLD agree within 0.07 dB (8.10/8.03/8.09 dB), because shared-scene tiles see near-IID data, so FedAvg is kept. On the NoC (Table IIIb), the Torus cuts parameter-server latency by 33% relative to a mesh, and RingAllReduce spreads the same 189.5 MB over all links, finishing in 18.96 ms with $22.7 \mu\text{J}$: $6.4\times$ faster and 56% less energy than a mesh parameter server. AllReduce and Gossip move $2.1\times$ and $2.7\times$ more bytes.

Honest cost accounting. At 600 samples per tile, FL moves $39.6\times$ more bytes than a one-time raw-CSI upload (Table IIIc), and it becomes cheaper only above $S^* \approx 23,752$ samples per tile. The case for Fed-RIS is therefore CSI locality, robustness, and latency: one forward pass replaces a 2.0–21.9 ms iterative solve that recurs in every coherence block, while training traffic is paid once.

Limitations. With perfect CSI, closed-form MRC is optimal and Fed-RIS trails it by 1.3 dB. The evaluation is limited to a single-antenna BS, a single user, and synthetic Rician channels.

VI. Conclusion

Fed-RIS federates RIS phase optimization across the tiles of one surface over an on-chip NoC. Raw CSI never leaves a tile, and the surface is configured in one forward pass per coherence block. On a 1024-element 28 GHz surface it recovers 36.7 dB of link budget, saves 31.4 dB of transmit power, and lies 1.3 dB from the perfect-CSI bound. It degrades most gracefully under CSI error (3.2 dB vs. 5.1 dB for AO/SCA), tolerates 2-bit shifters at 0.82 dB, and aggregates $6.4\times$ faster with 56% less energy on a Torus with RingAllReduce. Future work covers cross-tile coherence, multi-antenna and multi-user operation, ray-traced channels, and hardware prototyping.

References

- [1] M. Di Renzo et al., “Smart radio environments empowered by reconfigurable intelligent surfaces: How it works, state of research, and the road ahead,” *IEEE J. Sel. Areas Commun.*, vol. 38, no. 11, pp. 2450–2525, Nov. 2020.
- [2] E. Basar, M. Di Renzo, J. de Rosny, M. Debbah, M.-S. Alouini, and R. Zhang, “Wireless communications through reconfigurable intelligent surfaces,” *IEEE Access*, vol. 7, pp. 116753–116773, 2019.
- [3] T. S. Rappaport et al., “Wireless communications and applications above 100 GHz: Opportunities and challenges for 6G and beyond,” *IEEE Access*, vol. 7, pp. 78729–78757, 2019.
- [4] Q. Wu and R. Zhang, “Intelligent reflecting surface enhanced wireless network via joint active and passive beamforming,” *IEEE Trans. Wireless Commun.*, vol. 18, no. 11, pp. 5394–5409, Nov. 2019.
- [5] Q. Wu and R. Zhang, “Beamforming optimization for wireless network aided by intelligent reflecting surface with discrete phase shifts,” *IEEE Trans. Commun.*, vol. 68, no. 3, pp. 1838–1851, Mar. 2020.
- [6] H. Guo, Y.-C. Liang, J. Chen, and E. G. Larsson, “Weighted sum-rate maximization for reconfigurable intelligent surface aided wireless networks,” *IEEE Trans. Wireless Commun.*, vol. 19, no. 5, pp. 3064–3076, May 2020.
- [7] S. Boyd, N. Parikh, E. Chu, B. Peleato, and J. Eckstein, “Distributed optimization and statistical learning via the alternating direction method of multipliers,” *Found. Trends Mach. Learn.*, vol. 3, no. 1, pp. 1–122, 2011.
- [8] Z.-Q. Luo, W.-K. Ma, A. M.-C. So, Y. Ye, and S. Zhang, “Semidefinite relaxation of quadratic optimization problems,” *IEEE Signal Process. Mag.*, vol. 27, no. 3, pp. 20–34, May 2010.
- [9] C. Huang, R. Mo, and C. Yuen, “Reconfigurable intelligent surface assisted multiuser MISO systems exploiting deep reinforcement learning,” *IEEE J. Sel. Areas Commun.*, vol. 38, no. 8, pp. 1839–1850, Aug. 2020.
- [10] Y. Shen, Y. Shi, J. Zhang, and K. B. Letaief, “Graph neural networks for scalable radio resource management: Architecture design and theoretical analysis,” *IEEE J. Sel. Areas Commun.*, vol. 39, no. 1, pp. 101–115, Jan. 2021.
- [11] M. Eisen and A. Ribeiro, “Optimal wireless resource allocation with random edge graph neural networks,” *IEEE Trans. Signal Process.*, vol. 68, pp. 2977–2991, 2020.
- [12] P. Velicković, G. Cucurull, A. Casanova, A. Romero, P. Liò, and Y. Bengio, “Graph attention networks,” in *Proc. ICLR*, 2018.

- [13] B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. A. y Arcas, "Communication-efficient learning of deep networks from decentralized data," in Proc. AISTATS, 2017, pp. 1273–1282.
- [14] T. Li, A. K. Sahu, M. Zaheer, M. Sanjabi, A. Talwalkar, and V. Smith, "Federated optimization in heterogeneous networks," in Proc. MLSys, 2020.
- [15] S. P. Karimireddy, S. Kale, M. Mohri, S. Reddi, S. Stich, and A. T. Suresh, "SCAFFOLD: Stochastic controlled averaging for federated learning," in Proc. ICML, 2020, pp. 5132–5143.
- [16] K. Yang, T. Jiang, Y. Shi, and Z. Ding, "Federated learning via over-the-air computation," IEEE Trans. Wireless Commun., vol. 19, no. 3, pp. 2022–2035, Mar. 2020.
- [17] A. M. Elbir and S. Coleri, "Federated learning for channel estimation in conventional and RIS-assisted massive MIMO," IEEE Trans. Wireless Commun., vol. 21, no. 6, pp. 4255–4268, Jun. 2022.
- [18] L. Benini and G. De Micheli, "Networks on chips: A new SoC paradigm," Computer, vol. 35, no. 1, pp. 70–78, Jan. 2002.
- [19] W. J. Dally and B. Towles, Principles and Practices of Interconnection Networks. Morgan Kaufmann, 2004.
- [20] P. Patarasuk and X. Yuan, "Bandwidth optimal all-reduce algorithms for clusters of workstations," J. Parallel Distrib. Comput., vol. 69, no. 2, pp. 117–124, 2009.